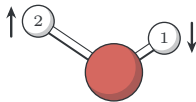
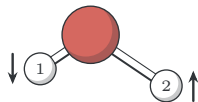


$$\xrightarrow[\chi_{\text{stat}}(\sigma) = -1]{\sigma = (12)}$$



$$\xrightarrow[\chi_{\pm}(E^*) = \pm 1]{E^*}$$


 $\mathbf{X}$

$$\Psi \downarrow$$

$$\begin{bmatrix} \uparrow & \downarrow \\ 1 & 2 \end{bmatrix}$$

 $\Psi(\mathbf{X})$

$$\xrightarrow{\sigma}$$

$$\sigma \mathbf{X} = \mathbf{X} \mathbf{P}_{\sigma}$$

$$\Psi \downarrow$$

$$-\begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix}$$

$$\Psi(\sigma \mathbf{X}) = \chi_{\text{stat}}(\sigma) \sigma \cdot \Psi(\mathbf{X})$$

$$\xrightarrow{E^*}$$

$$E^* \sigma \mathbf{X} = -\mathbf{X} \mathbf{P}_{\sigma}$$

$$\Psi \downarrow$$

$$\mp \begin{bmatrix} \downarrow & \uparrow \\ 1 & 2 \end{bmatrix}$$

$$\Psi(E^* \sigma \mathbf{X}) = \chi_{\pm} \Psi(\sigma \mathbf{X})$$